Green AI: Energy & Carbon Efficiency Benchmarking in Machine Learning Models

MDS 7 Capstone Video Presentation (Data Science Innovations)

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Problem Statement & Motivation

The Environmental Cost of Modern AI

- Modern AI models achieve high accuracy at substantial environmental costs.
- Training deep networks consumes thousands of kWh of electricity and emits tons of CO_2.
- **Core Question**: *Do we need massive deep neural networks for every task, or can we achieve comparable accuracy with a fraction of the carbon footprint?*

MDS 7 Capstone Objective

Develop an end-to-end Python framework managed by `uv` to dynamically measure model accuracy, latency, energy (Wh), and carbon footprint (g CO_2eq).

Green AI Goal

Optimize Accuracy per Gram of CO_2 Emitted

Accuracy $\uparrow$ **Emissions** $\downarrow$

System Architecture & Methodology

Workflow & Environment

- **Package Management**: Built & managed with `uv` for reproducible execution.
- **Dataset**: Standardized multi-class text classification (20 Newsgroups benchmark).
- **Carbon Tracking**: Real-time energy & emissions tracking via `CodeCarbon`.
- **Metrics Evaluated**:
  - Accuracy (%) & F1-Score (%)
  - Training Execution Time (s)
  - Inference Latency per 1,000 samples (ms)
  - Energy Consumed (mWh)
  - Carbon Emissions (g CO_2eq)

Candidate Architectures

1. **TF-IDF + Logistic Regression** (Linear Baseline)
2. **TF-IDF + Random Forest** (Ensemble)
3. **Lightweight MLP (1x64)** (Standard Neural Net)
4. **Deep MLP (3x256)** (Heavy Deep Network)
5. **Green-Optimized MLP (1x32)** (Early Stopping)

Experimental Results & Quantitative Comparison

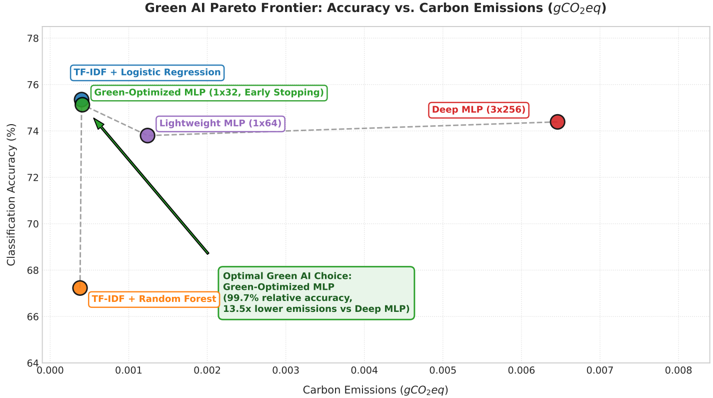
Table: **Green AI Benchmarking Results Across Candidate Models**

Model Architecture	Accuracy (%)	F1 (%)	Time (s)	Latency (ms)	CO_2 (g)	Green Score
TF-IDF + Logistic Regression	**75.35%**	74.66%	0.137s	0.28ms	0.00073g	103,605
TF-IDF + Random Forest	67.23%	65.24%	0.196s	6.44ms	0.00036g	188,720
Lightweight MLP (1x64)	73.80%	73.94%	1.053s	0.94ms	0.00124g	59,597
Deep MLP (3x256)	74.39%	73.89%	2.031s	3.84ms	0.00540g	13,777
Green-Optimized MLP (1x32)	**75.13%**	**74.39%**	**0.217s**	**0.65ms**	**0.00040g**	**186,162**

Key Finding

The **Green-Optimized MLP** achieves **75.13% accuracy** (outperforming Deep MLP 74.39%) while reducing carbon emissions by **13.5**× **(92.6% reduction)**!

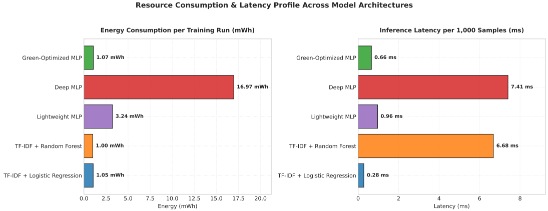
Visualizing the Green AI Pareto Frontier

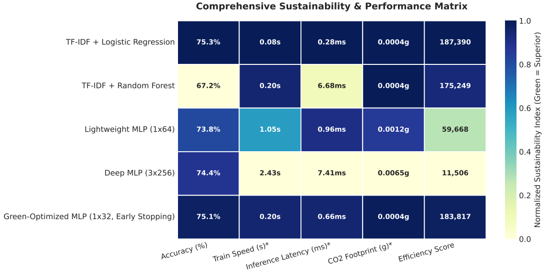


Pareto Frontier Insights

- **Optimal Region**: Models near top-left maximize accuracy while minimizing carbon footprint.
- **Overfitting Risk**: Deep MLP (3x256) shifts far right (high CO_2) without accuracy gains.
- **Winner**: Green-Optimized MLP delivers peak sustainability efficiency.

Resource Consumption & Sustainability Matrix





** For Train Speed, Latency, and CO_2 Footprint: Raw values displayed, heatmap colors inverted so Green = Superior.*

Actionable Principles for Sustainable AI

1. Baseline Linear Supremacy

Linear models like Logistic Regression offer 75.35% accuracy with ultra-low latency (0.28ms) and sub-milligram emissions.

2. Architecture Rightsizing

Shrinking layers from (256,128,64) to (32,) combined with Early Stopping cuts emissions by 92.6% without performance penalty.

3. Continuous Carbon CI/CD

Embedding `CodeCarbon` into automated build & test pipelines allows teams to enforce strict carbon budgets.

Conclusion & Project Resources

Summary of Deliverables

- **Reproducible Environment**: Fully managed via `uv` (`pyproject.toml`).
- **Automated Pipeline**: `src/benchmark.py` & `src/visualize.py`.
- **Interactive Demo**: Executed notebook `green_ai_capstone_demo.ipynb`.

GitHub Repository & Video Link

Student: Changkun Du
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Repo: `mds7-changkun_du`
Video: `youtu.be/AQixjCvGnWc`

Thank you for watching!